

Panamex Import Management | Container & Demurrage OS

Enterprise Logistics Control Tower, Automated Free-Time Calculator & Port Demurrage Prevention System

CLIENT ORGANIZATION

Panamex Pacific (PNG) Ltd

OPERATING PORTS

Lae & Port Moresby

SOLUTION SCOPE

Turnkey Web Platform

DEPLOYMENT LEAD TIME

4 Weeks Turnaround

1. EXECUTIVE SUMMARY & PROBLEM ANALYSIS

Panamex Pacific (PNG) Ltd receives large volumes of containerized FMCG cargo through Papua New Guinea's primary maritime ports (Lae Tidal Basin & Port Moresby Motukea Port). Shipping lines (Swire, MSC, PIL, Mariana Express, ANL) enforce strict free-time periods (normally 14 days), following which punitive demurrage charges accrue daily (\$120 – \$250+ per TEU/day).

In addition, PNG Ports allows only **5 Free Days** from vessel full discharge to port clearance before wharfage penalty rates escalate. Delays in customs approvals, NAQIA PTL certification, wharfage payments, Delivery Orders (DO), or Gate Pass (GP) issuance directly trigger severe financial losses.

Financial Risk & Business Value: Preventing demurrage on just **3 to 4 containers each month** saves Panamex an estimated **\$25,000 – \$45,000+ (PGK 95,000 – 170,000+) annually**, providing complete Return on Investment (ROI) within **60 to 75 days** of deployment.

2. SYSTEM CAPABILITIES & ARCHITECTURE DELIVERED

- **Executive Command Dashboard:** 5-stage interactive Live Cargo Movement Pipeline (Sea Transit → Port Wharf → Formalities → Yard Destuffing → Returned Empty) with live percentage completion indicators.
- **Demurrage & Port Free-Days Automation:** Automatic derivation of Due Dates, 5-day wharf limits, and smart status flags (Within Free Time, Port Due Soon, Overdue, Returned).
- **Pre-Clearance Formalities Funnel:** Milestone checklist monitoring Customs Draft Approval, NAQIA PTL clearance, Wharfage & Handling payment, Shipping DO, Gate Pass, and Gate-out.
- **Real-Time Yard Operations:** Real-time PNG Operations Clock (UTC+10:00) with depot allocation tracking across Panamex Head Depot (1B) and Agent Yard (6A).
- **One-Click Compliance Reporting:** Instant client-side Excel (.xlsx) and high-res executive PDF (.pdf) generator for audit reviews and shipping line reconciliations.
- **Enterprise Security & Audit Trail:** Multi-tier RBAC (Admin, Shipping Manager, Yard Staff) with tamper-proof diff inspection logging every container edit and deletion.
- **Dual Theme UI:** High-contrast Obsidian Dark Theme and Daylight Light Theme with ambient depth.

3. COMPREHENSIVE COMMERCIAL PROPOSAL & COST MATRIX

The cost structure is segmented into **One-time Implementation**, **Cloud Infrastructure & Hosting**, **Security & Data Protection**, and **Ongoing Support / Maintenance (SLA)**.

ITEM	COMPONENT & TECHNICAL SCOPE	DELIVERABLES INCLUDED	BILLING	COST (USD)	COST (PGK)
1.0	Custom Software Development Core business logic, reactive UI, KPI diagrams, demurrage engine, bulk upload, Excel/PDF generators, dark/light themes.	Complete turnkey application, source code, tailored PNG port rules	One-Time	\$4,800	K 18,250
2.0	Cloud Server & Hosting Setup Dedicated high-speed cloud VPS provisioning, custom domain binding, Node.js production daemon runtime, local storage engine.	Server provisioning, DNS, firewall, SSL TLS 1.3 setup	One-Time	\$450	K 1,700
3.0	Enterprise Security & Compliance Hardening Cloudflare WAF / Anti-DDoS, RBAC authentication, input sanitization, automated encrypted offsite backup system.	Security hardening, automated daily backups, penetration testing	One-Time	\$650	K 2,450
4.0	User Training, Testing & Commissioning UAT testing, operational user manuals, onboarding sessions for Admin, Shipping Lead, and Yard staff.	Training docs, operator onboarding, go-live cutover	One-Time	\$600	K 2,300
TOTAL ONE-TIME IMPLEMENTATION CHARGES (Turnkey Deployment)				\$6,500	K 24,700

4. RECURRING OPERATIONAL EXPENSES (ANNUAL HOSTING, SECURITY & AMC)

ITEM	OPERATIONAL COMPONENT	SPECIFICATIONS & INCLUSIONS	ANNUAL (USD)	ANNUAL (PGK)
A.1	Managed Cloud Server Hosting 4GB RAM, 2 vCPU SSD Cloud VPS, 99.9% uptime SLA, unmetered bandwidth.	Dedicated cloud instance, OS patching, 24/7 uptime monitoring	\$720 / yr (\$60/mo)	K 2,750 / yr
A.2	Security, SSL & Automated Backups Cloudflare Pro WAF, DDoS protection, wildcard SSL certificate, encrypted offsite backups.	Daily automated snapshots, 30-day retention, disaster recovery	\$380 / yr (\$32/mo)	K 1,450 / yr
A.3	Annual Maintenance Contract (AMC & SLA) Priority tech support, bug resolution, browser compatibility, minor adjustments, performance audits.	Business-hours priority SLA, dedicated developer assistance	\$1,400 / yr (\$117/mo)	K 5,300 / yr
TOTAL RECURRING ANNUAL OPERATIONAL EXPENSES			\$2,500 / yr	K 9,500 / yr

5. INVESTMENT PACKAGES FOR MANAGEMENT DECISION

RECOMMENDED		
Tier 1: Standard \$4,500 K 17,100 One-time \$950/yr cloud	Tier 2: Professional Suite \$6,500 K 24,700 One-time \$2,500/yr suite	Tier 3: Enterprise Cloud \$8,900 K 33,800 One-time \$3,400/yr suite
✓ Core container tracking ✓ Demurrage calculation engine ✓ Basic Excel export ✓ Single-tier user access ✓ 60 days warranty support	✓ Full Executive Analytics & Pipeline ✓ Real-time Yard Tracker (1B & 6A) ✓ Excel & Landscape PDF Export ✓ Role-Based Access (Admin/Mgr/Staff) ✓ Tamper-proof Diff Audit Trail ✓ Dark & Light Theme Experience ✓ Cloudflare WAF & Daily Backups ✓ 1 Year Priority SLA & AMC included	✓ Everything in Professional Suite ✓ Multi-branch Port segregation ✓ Automated SMS/Email Gateway ✓ ERP / SAP Integration hooks ✓ 24/7 Dedicated Emergency SLA ✓ Semi-annual feature expansions

6. IMPLEMENTATION TIMELINE & PAYMENT SCHEDULE

Delivery Milestones (4 Weeks):

- **Week 1:** Cloud Server provisioning, theme engine, database schema.
- **Week 2:** Demurrage engine, PNG ports wharf logic, bulk manifest upload.
- **Week 3:** Visual analytics, gauge meters, Excel/PDF reporting, audit diffs.
- **Week 4:** Security penetration testing, operator training, final go-live.

Commercial Payment Milestones:

- **40% Deposit:** Mobilization & infrastructure provisioning.
- **40% Interim:** Successful demonstration on staging server.
- **20% Final:** Official handover, user training, and production cutover.

7. AUTHORIZATION & SIGN-OFF

To accept this proposal and initiate Phase 1 deployment under the selected tier:

PREPARED & SUBMITTED BY Solution Architect / Development Lead Panamex Technology & Logistics Systems Date: September 14, 2026	ACCEPTED & APPROVED BY (CLIENT) Authorized Signatory / Executive Management Panamex Pacific (PNG) Ltd Date: _____
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